

APPENDIX/SUPPLEMENTARY MATERIALS: CAMIE: CONTEXT-AWARE SUBGRAPH EXPLANATIONS FOR MULTI-TASK GNNs

A Detailed Experimental Results

Table 1: Per-assay Fidelity $_{F1}^+$ comparison between ST-MIE and MISA on selected assays, sorted by sample size. Full results are reported in Appendix.

Group	Assay	#Samples	MISE	ST-MIE
Low	NVS_NR_rMR	77	0.7586 ± 0.2686	0.6689 ± 0.3429
	NVS_ADME_hCYP1A1	137	0.3095 ± 0.2319	0.2746 ± 0.2077
	NVS_NR_hTRa_Antagonist	187	0.2632 ± 0.0827	0.2430 ± 0.0499
	NVS_NR_hAR	189	0.4599 ± 0.2464	0.4820 ± 0.2291
	NVS_ADME_hCYP19A1	198	0.4447 ± 0.1395	0.4035 ± 0.1734
	NVS_NR_rAR	223	0.3678 ± 0.2414	0.3680 ± 0.2253
	NVS_NR_cAR	248	0.5676 ± 0.3267	0.5039 ± 0.3463
	NVS_NR_hGR	282	0.2642 ± 0.2367	0.2375 ± 0.2237
	CEETOX_H295R_OHPREG_up	364	0.4270 ± 0.0822	0.4472 ± 0.0825
	CEETOX_H295R_PROG_up	371	0.4227 ± 0.1156	0.4088 ± 0.1084
Avg.			$0.4285(+0.0248)$	0.4037
Medium	CEETOX_H295R_OHPROG_up	390	0.4290 ± 0.1168	0.4250 ± 0.1028
	CEETOX_H295R_OHPROG_dn	390	0.3382 ± 0.2147	0.2989 ± 0.2323
	CEETOX_H295R_ESTRONE_up	390	0.1976 ± 0.1581	0.2166 ± 0.1490
	CEETOX_H295R_ESTRONE_dn	390	0.1859 ± 0.1251	0.1626 ± 0.1368
	CEETOX_H295R_CORTISOL_dn	390	0.4547 ± 0.1830	0.4054 ± 0.2061
	CEETOX_H295R_11DCORT_dn	390	0.3268 ± 0.2400	0.2833 ± 0.2411
	NVS_NR_hER	945	0.3752 ± 0.1350	0.3591 ± 0.1327
	OT_AR_ARSRC1_0960	1461	0.1793 ± 0.1166	0.1445 ± 0.1490
	OT_ER_ERbERb_0480	1467	0.4484 ± 0.0569	0.4269 ± 0.0807
	OT_ER_ERaERb_0480	1477	0.4808 ± 0.0503	0.4714 ± 0.0525
Avg.			$0.3416 (+0.0222)$	0.3194
High	ATG_GR_TRANS_up	2938	0.1367 ± 0.1034	0.1282 ± 0.0995
	ATG_GR_TRANS_dn	2938	0.0000 ± 0.0000	0.0000 ± 0.0000
	ATG_ERa_TRANS_up	2938	0.3985 ± 0.0684	0.3762 ± 0.0845
	ATG_AR_TRANS_dn	2938	-0.0119 ± 0.0315	-0.0104 ± 0.0295
	ATG_AR_TRANS_up	2938	0.3027 ± 0.0857	0.2945 ± 0.0840
	ATG_ERE_CIS_dn	2938	-0.0156 ± 0.0413	-0.0125 ± 0.0354
	ATG_ERE_CIS_up	2938	0.3507 ± 0.0805	0.3258 ± 0.1056
	TOX21_VDR_BLA_agonist_ratio	5872	-0.0123 ± 0.0273	-0.0124 ± 0.0254
	TOX21_TR_LUC_GH3_Antagonist	6353	0.1978 ± 0.1724	0.1467 ± 0.2152
	TOX21_AR_LUC_MDAKB2_Agonist	6353	0.4470 ± 0.0585	0.4018 ± 0.1369
Avg.			$0.1793 (+0.0156)$	0.1638

Table 2: Top- k sensitivity analysis. Values are reported as mean $\pm$ std across seeds.

Model	k	Fidelity $_{S2}$ ($\uparrow$)	Fidelity $_{S1}$ ($\downarrow$)
MISE (ours)	1	0.2774 $\pm$ 0.0899	0.1593 $\pm$ 0.0610
	2	0.3024 $\pm$ 0.0924	0.0945 $\pm$ 0.0458
	3	0.3076 $\pm$ 0.0900	0.0554 $\pm$ 0.0373
	4	0.3078 $\pm$ 0.0884	0.0369 $\pm$ 0.0337
	5	0.3077 $\pm$ 0.0877	0.0228 $\pm$ 0.0244
eXEL	1	0.2635 $\pm$ 0.0883	0.1617 $\pm$ 0.0609
	2	0.2913 $\pm$ 0.0918	0.1053 $\pm$ 0.0466
	3	0.2956 $\pm$ 0.0901	0.0751 $\pm$ 0.0375
	4	0.2985 $\pm$ 0.0879	0.0601 $\pm$ 0.0324
	5	0.2987 $\pm$ 0.0880	0.0470 $\pm$ 0.0272
SA	1	0.1358 $\pm$ 0.0458	0.1756 $\pm$ 0.0777
	2	0.1861 $\pm$ 0.0712	0.1540 $\pm$ 0.0728
	3	0.2178 $\pm$ 0.0880	0.1235 $\pm$ 0.0630
	4	0.2432 $\pm$ 0.1012	0.0981 $\pm$ 0.0541
	5	0.2582 $\pm$ 0.1048	0.0698 $\pm$ 0.0459
GBP	1	0.1685 $\pm$ 0.0307	0.1913 $\pm$ 0.0572
	2	0.2405 $\pm$ 0.0546	0.1622 $\pm$ 0.0505
	3	0.2746 $\pm$ 0.0685	0.1206 $\pm$ 0.0415
	4	0.2924 $\pm$ 0.0779	0.0888 $\pm$ 0.0365
	5	0.3018 $\pm$ 0.0828	0.0611 $\pm$ 0.0343
GradCAM	1	0.1910 $\pm$ 0.0323	0.1781 $\pm$ 0.0616
	2	0.2455 $\pm$ 0.0534	0.1437 $\pm$ 0.0530
	3	0.2723 $\pm$ 0.0664	0.1052 $\pm$ 0.0438
	4	0.2864 $\pm$ 0.0753	0.0739 $\pm$ 0.0355
	5	0.2942 $\pm$ 0.0784	0.0512 $\pm$ 0.0325
PGExplainer	1	0.1654 $\pm$ 0.0641	0.1930 $\pm$ 0.0639
	2	0.2323 $\pm$ 0.0913	0.1407 $\pm$ 0.0551
	3	0.2623 $\pm$ 0.0929	0.1041 $\pm$ 0.0444
	4	0.2767 $\pm$ 0.0919	0.0765 $\pm$ 0.0334
	5	0.2874 $\pm$ 0.0907	0.0559 $\pm$ 0.0262
GNExplainer	1	0.1045 $\pm$ 0.0194	0.2091 $\pm$ 0.0637
	2	0.1566 $\pm$ 0.0385	0.1912 $\pm$ 0.0637
	3	0.2006 $\pm$ 0.0515	0.1639 $\pm$ 0.0591
	4	0.2362 $\pm$ 0.0627	0.1337 $\pm$ 0.0494
	5	0.2596 $\pm$ 0.0705	0.1046 $\pm$ 0.0416

Algorithm 1 Training and Motif Ranking of CAMIE

Require: Training dataset $\mathcal{D}$; query pair (G, a_j) ; motif sets $\{\mathcal{M}_{G^{(i)}}\}_{i=1}^N$ and $\mathcal{M}_G$; motif partitions $\{\mathcal{P}_{G^{(i)}}\}_{i=1}^N$ and $\mathcal{P}_G$; frozen encoder f_ϕ ; readout function r ; frozen output head π ; assay-context encoder $g(\cdot; \eta)$; context-conditioned motif scorer q_θ ; number of selected motifs k

Ensure: Top- k context-aware motif explanation $\mathcal{E}_k(G, a_j)$

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1: for  $i = 1$  to  $N$  do
2:    $H_{G^{(i)}} \leftarrow f_\phi(X^{(i)}, A^{(i)})$ 
3:    $z_{G^{(i)}} \leftarrow r(H_{G^{(i)}})$ 
4:   for all  $m_s \in \mathcal{M}_{G^{(i)}}$  with node index set  $I_s \in \mathcal{P}_{G^{(i)}}$  do
5:      $z_{m_s}^{(i)} \leftarrow \frac{1}{|I_s|} \sum_{v \in I_s} h_v^{(i)}$ 
6:     Define  $X_{\setminus m_s}^{(i)}$  by  $(X_{\setminus m_s}^{(i)})_v \leftarrow \mathbf{1}[v \notin I_s] X_v^{(i)}$ 
7:      $H_{G^{(i)} \setminus m_s} \leftarrow f_\phi(X_{\setminus m_s}^{(i)}, A^{(i)})$ 
8:      $z_{G^{(i)} \setminus m_s} \leftarrow r(H_{G^{(i)} \setminus m_s})$ 
9:     for all  $a_j \in \mathcal{T}^{(i)}$  do
10:       $z_{a_j} \leftarrow g(a_j; \eta)$ 
11:       $\tilde{z}_{a_j} \leftarrow W_a z_{a_j} + b_a$ 
12:       $u_{s,j}^{(i)} \leftarrow [z_{m_s}^{(i)}; z_{G^{(i)}}; \tilde{z}_{a_j}; z_{m_s}^{(i)} \odot z_{G^{(i)}}; z_{m_s}^{(i)} \odot \tilde{z}_{a_j}]$ 
13:       $\hat{t}_{s,j}^{(i)} \leftarrow q_\theta(u_{s,j}^{(i)})$ 
14:       $t_{s,j}^{(i)} \leftarrow |\pi(z_{G^{(i)}})_j - \pi(z_{G^{(i)} \setminus m_s})_j|$ 
15:    end for
16:  end for
17: end for
18:  $\theta^* \leftarrow \arg \min_\theta \mathcal{L}(\mathcal{D}; \theta)$ 
19: Compute  $H_G \leftarrow f_\phi(X, A)$ ,  $z_G \leftarrow r(H_G)$ , and  $z_{a_j} \leftarrow g(a_j; \eta)$ 
20: for all  $m_s \in \mathcal{M}_G$  with node index set  $I_s \in \mathcal{P}_G$  do
21:    $z_{m_s} \leftarrow \frac{1}{|I_s|} \sum_{v \in I_s} h_v$ 
22:    $\tilde{z}_{a_j} \leftarrow W_a z_{a_j} + b_a$ 
23:    $u_{s,j} \leftarrow [z_{m_s}; z_G; \tilde{z}_{a_j}; z_{m_s} \odot z_G; z_{m_s} \odot \tilde{z}_{a_j}]$ 
24:    $\hat{t}_{s,j} \leftarrow q_{\theta^*}(u_{s,j})$ 
25: end for
26:  $\mathcal{E}_k(G, a_j) \leftarrow \text{Top}_k(\{(m_s, \hat{t}_{s,j})\}_{s=1}^M)$ 
27: return  $\mathcal{E}_k(G, a_j)$ 

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